



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY
MALAWI INSTITUTE OF TECHNOLOGY
APPLIED STUDIES DEPARTMENT

2026 MID-SEMESTER ONE EXAMINATIONS

MATH3115: COMPLEX ANALYSIS

DATE: Wednesday, 25th March, 2026.

TIME : 13:30 - 16:30

EXAMINER: M. Kanyika

Instructions

1. This paper contains 3 printed pages, including the cover page. Please check.
2. Attempt **all 6** questions.
3. Please write clearly, legibly and show all your working.
5. Write down your **Registration number** at the top of each page of the answer sheet.
6. Use of non-programmable calculators is allowed.

Question	1	2	3	4	5	6	Total
Marks	7	10	13	10	14	26	80

DO NOT TURN OVER THE QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

Question 1

- (a) Let $z = 2 + 3i$ $w = 2 + 7i$ $v = 1 + i$, perform the following
- i. $w + 2v$ (3 marks)
 - ii. $2z - 3w$ (4 marks)

Question 2

- (a) Given that $z = 3 - i$, $w = 5 + 2i$ and $v = 2 + 2i$, compute the following:
- i. $z - w$ (4 marks)
 - ii. zv (6 marks)

Question 3

- (a) Find the value of $(\cos 7^\circ + i \sin 7^\circ)^{30}$. (3 marks)
- (b) Perform the required calculation and express the answer in the form of $a + bi$.
- i. $(3 - i) - i(4 + 5i)$ (5 marks)
 - ii. $(1 - i)^2$ (5 marks)

Question 4

- (a) Simplify $\frac{3-5i}{2+9i}$. (5 marks)
- (b) Let $z_1 = 2 + 5i$ and $z_2 = 7 - 3i$, be two complex numbers. Verify if the commutativity of addition hold for z_1 and z_2 . (5 marks)

Question 5

- (a) Consider $z = -5 + 12i$. Find the modulus of z (4 marks)
- (b) Evaluate $\int_0^\pi \sinh(it) dt$. (10 marks)

Question 6

- (a) Use binomial Expansion to show that

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

(6 marks)

- (b) Find the cube root of $-16 + 16i$ (10 **marks**)
- (c) Find a parametrization of the polygonal path $C = C_1 + C_2 + C_3$ from start to the end using three straight line segments, given the points:

$$z_0 = -2 + i, \quad z_1 = -2 - 2i, \quad z_2 = 1 - 2i \quad z_3 = 3 + i$$

(10 **marks**)

END OF QUESTION PAPER